Reader Supplement to

Triangle Hypersurfaces and a Sharp Identifiability Boundary for Level-2 K3P Networks

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This supplement is a reader-facing map of the exact proof package. It gives the formulas, finite censuses, witness selections, and hash bindings needed to audit the article without reproducing raw JSON ledgers. All paths are relative to the project root `k3p_level2_identifiability_final/`. The machine-readable files, not line-broken values in this document, are canonical for exact replay.

1 Dependency graph

The necessity and sufficiency chains are:

```
K3P domain + root invariance
  -> true-cut rank <= 4 + generic noncut recovery
  -> Cut(target) subset Cut(source)
  -> crossing-split dichotomy + 204 pointwise one-active directions
  -> Cut(source) subset Cut(target)
  -> common labelled reduced bridge tree
  -> three-sector bridge fibre + physical local product
  -> marginal submersion + finite target-type localization
  -> 14-orbit theorem + restoration + coherent probes
  -> necessity: topology equivalence modulo ordinary triangles

H14 quartic + three rank-14 orientation sections
  -> common relative triangle germ
  -> one labelled contextual contraction
  -> simultaneous principal/CT bridge gluing
  -> sufficiency and symmetric common full-dimensional germ

main classification + finite topology set + semialgebraic stratification
  -> proper exceptional set
  -> generic identifiability + terminating exact reconstruction

weak-not-strong topology census + Krawczyk equality box + two rank-15 maps
  -> base ambient-open common germ
  -> six-dimensional identical-cherry inverse
  -> all-n dimension 6n-3

all active theorem branches + final claim lock
  -> integrated K3P-SAME classification gate
  -> 16/16 fail-closed integrated mutations
  -> CERTIFIED_K3P_SAME mathematics; publication engineering remains separate
```

The cut-transfer chain uses neither a common bridge tree nor the fourteen-orbit classification. The local classification is invoked only after cut equality has been proved.

2 Theorem-to-certificate crosswalk

claim	minimal active evidence and replay boundary
K3P inverse Fourier formulas, $\mathcal{D}_{3,+}, \mathcal{D}_{3,\mathrm{CT}}$, subdivision, root movement	`model_domain/primary_exact_evidence.json`; primary exact replay.
Cycle–theta primitive topology, no-omnian grammar, and strong repairs	`input_frozen/referenced_chat_manuscripts/jc_level2_source.tex` and `input_frozen/referenced_chat_manuscripts/k2p_level2_source.tex`, with the graph-only import boundary fixed by `COMPANION_DEPENDENCY_LOCK.json` and independently rebuilt in the clean-room, restoration, and probe replays.
True-cut rank and generic noncut recovery	article derivation; frozen JC cut theorem only on the relatively open strict JC locus in the isotropic slice, not on an open subset of full K3P; `cut_recovery/adversarial/` preserves the failed universal shortcut.
204 pointwise one-active directions	`cut_recovery/strong_crossbridge/final_certificate/STRONG_CROSSBRIDGE_FINAL_CERTIFICATE.json` and its producer-independent verifier.
Strong-class cut-set equality under containment	`cut_recovery/strong_crossbridge/global_transfer/THEOREM_MANIFEST.json` and `cut_recovery/strong_crossbridge/global_transfer/GLOBAL_TRANSFER_AUDIT.md`; independent adversarial replay.
Tree–sunlet separation and triangle rank	`three_port/primary_exact_evidence.json`, `input_frozen/k3p_cloud_artifacts/k3p_tree_sunlet_separator.json`.
H_{14}, irreducibility, smooth common germ, contextualization	`triangle_h14/K3P_H14_CONTEXT_CERTIFICATE.json`.
Three-sector bridge fibre and physical product	`bridge_fibre/K3P_BRIDGE_FIBRE_CERTIFICATE.json`.
Marginal descriptors and physical submersions	`marginals/K3P_MARGINAL_SUBMERSION_CERTIFICATE.json`.
Fourteen orbits and two sink swaps	`four_port_atlas/primary_exact_evidence.json`; `clean_room/H21_01_TRANSPORT_AUDIT.json`.
Fixed-full restoration	`restoration/RESTORATION_MANIFEST.json`, `restoration/K3P_RESTORATION_THEOREM_REPORT.md`, ledger, proof registry, `restoration/K3P_RESTORATION_INDEPENDENT_VERIFICATION.json`, and `restoration/K3P_RESTORATION_MUTATION_CERTIFICATE.json`; the independent replay passes and all 20 fail-closed restoration mutations are rejected.
One-/two-port coherence	`probes/K3P_PROBE_COHERENCE_CERTIFICATE.json`, independent streaming replay, mutation certificate, and the v2 K3P restoration binding `probes/RESTORATION_MANIFEST.json`.
Global gluing, genericity, reconstruction	`global_infrastructure/K3P_GLOBAL_GLUE_AND_RECONSTRUCTION_CERTIFICATE.json`, `global_infrastructure/GLOBAL_INFRASTRUCTURE_MANIFEST.json`, and its independent replay, read together with the active cut-transfer and restoration bindings. The integrated primary gate passes all 28 entries.
Final K3P-SAME classification promotion	`reproducibility/K3P_SAME_CLASSIFICATION_GATE_REPORT.json` has schema `k3p-same-integrated-classification-gate-v2`, status `CERTIFIED_K3P_SAME`, and no remaining mathematical gates; `reproducibility/K3P_SAME_CLASSIFICATION_MUTATION_REPORT.json` rejects all 16 integrated fail-closed mutations.

claim	minimal active evidence and replay boundary
Weak-class sharpness and all n	`sharpness/K3P_SHARPNESS_KRAWCZYK_CERTIFICATE.json` and `sharpness/K3P_SHARPNESS_TOPOLOGY_ALL_N_CERTIFICATE.json`.
Outer tree–double-theta obstruction, including strict-CT deformation	`input_frozen/referenced_chat_manuscripts/tree_theta_collision_source.tex` contains the exact tangent and analytic implicit-function argument; `topology/primary_double_theta_evidence.json` and `topology/primary_rooting_census_evidence.json` bind the collision rank, dimension, and exclusion from weak tree-childness in the active primary replay.

3 Exact model domains

For an edge spectrum $(1, c, g, t)$, inverse Fourier transformation gives

$$\frac{1}{4}(1+c+g+t), \quad \frac{1}{4}(1+c-g-t), \quad \frac{1}{4}(1-c+g-t), \quad \frac{1}{4}(1-c-g+t).$$

Thus

$$\mathcal{D}_{3,+} = \{(c,g,t) \in (0,1)^3 : 1+c-g-t > 0,\ 1-c+g-t > 0,\ 1-c-g+t > 0\}.$$

Strict continuous time is

$$\mathcal{D}_{3,\mathrm{CT}} = \{(c,g,t) \in (0,1)^3 : c > gt,\ g > ct,\ t > cg\}.$$

For strict subdivision, choose

$$r > \max\{c, g, t, g+t-c, c+t-g, c+g-t\}$$

in the principal domain, or additionally $r > \max\{gt/c, ct/g, cg/t\}$ in continuous time, and use

$$(c,g,t) = (r,r,r) \odot (c/r, g/r, t/r).$$

For a serial class the selected descriptor is

$$((c_i, g_i, t_i))_{i=1}^m \mapsto (\prod_i c_i, \prod_i g_i, \prod_i t_i).$$

Its Jacobian has three disjoint positive rows and rank three.

4 The H_{14} coordinate chart and quartic

The conservation-supported coordinate order is

$$\begin{aligned}(&000, 0CC, 0GG, 0TT, C0C, CC0, CGT, CTG,\\ &G0G, GCT, GG0, GTC, T0T, TCG, TGC, TT0).\end{aligned}$$

With $q_{000} = 1$, the ambient chart is $\mathbb{A}^{15}$. The exact quartic is

$$\begin{aligned}F_{H_{14}} = {} & q_{000}q_{CGT}q_{GTC}q_{TCG} - q_{000}q_{CTG}q_{GCT}q_{TGC}\\ & - q_{0CC}q_{CGT}q_{G0G}q_{TT0} + q_{0CC}q_{CTG}q_{GG0}q_{T0T}\\ & + q_{0GG}q_{C0C}q_{GCT}q_{TT0} - q_{0GG}q_{CC0}q_{GTC}q_{T0T}\\ & - q_{0TT}q_{C0C}q_{GG0}q_{TCG} + q_{0TT}q_{CC0}q_{G0G}q_{TGC}.\end{aligned}$$

The common tensor in this order is

$$\left(1, \tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{48}, \tfrac{1}{48}, \tfrac{1}{12}, \tfrac{1}{48}, \tfrac{1}{12}, \tfrac{1}{48}, \tfrac{1}{12}, \tfrac{1}{48}, \tfrac{1}{48}, \tfrac{1}{12}\right).$$

At this point $F = 0$ and six normalized gradient coordinates are nonzero, each with absolute value 1/6912. The irreducibility certificate treats the quartic as linear in q_{0CC}, with coefficient $-q_{CGT}q_{G0G}q_{TT0} + q_{CTG}q_{GG0}q_{T0T}$; a specialization kills that coefficient but leaves remainder -1.

5 Triangle rank witnesses

All three common preimages have edge order (a, b, c, d, e, f), spectra $(1/2, 1/2, 1/2)$ on the first five edges, $(1/3, 1/3, 1/3)$ on f, and inheritance 1/2.

orientation	normalized output rows	parameter columns	determinant
1	$(1, 2, 3, 4, 6, 8, 7, 10, 12, 9, 13, 5, 11, 15)$	$(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 16, 17)$	$-1/760840571584512$
2	$(4, 8, 12, 5, 7, 3, 6, 1, 10, 9, 11, 2, 13, 15)$	same	$-1/760840571584512$
3	$(4, 8, 12, 5, 6, 3, 7, 1, 10, 11, 9, 2, 13, 15)$	same	$1/760840571584512$

The tree rank-9 witness uses output rows $(4, 8, 12, 5, 6, 3, 7, 1, 10)$, parameter columns $0, \ldots, 8$, and determinant 8192/28940625.

6 Tree–sunlet six-circuit certificate

Write $I(L; R) = \prod_{u\in L} q_u - \prod_{v\in R} q_v$. The six circuits and their non-arm factors after substitution into the third-leaf sunlet map are:

	$L; R$	non-arm factor (up to the displayed nonzero sign and inheritance/arm monomial)
I_1	$000, CGT, GTC; 0TT, C0C, GG0$	$(d_C d_T - d_G)(d_G e_T f_C - e_C f_T)$
I_2	$000, CTG, TGC; 0GG, C0C, TT0$	$(d_C d_G - d_T)(d_T e_G f_C - e_C f_G)$
I_3	$000, GCT, TGC; 0CC, GG0, T0T$	$(d_C d_T - d_G)(d_G e_C f_T - e_T f_C)$
I_4	$000, GTC, TCG; 0CC, G0G, TT0$	$(d_C d_G - d_T)(d_T e_C f_G - e_G f_C)$
I_5	$000, CTG, GCT; 0TT, CC0, G0G$	$(d_C - d_G d_T)(d_C e_T f_G - e_G f_T)$
I_6	$000, CGT, TCG; 0GG, CC0, T0T$	$(d_C - d_G d_T)(d_C e_G f_T - e_T f_G)$

All six vanish on a tree. If all vanished on a strict sunlet, then, sectorwise, every nonzero composition margin has a pair of circuits whose two reciprocal cross equations force $d_h^2 = 1$. This contradicts $0 < d_h < 1$. If all three composition margins vanish instead, then $p = d_C d_G d_T$ satisfies $p = p^2$, again impossible for $p \in (0, 1)$. Hence $\sum_{j=1}^{6} I_j^2 > 0$.

7 Three-sector incidence normalizers

For an unmarked component of degree $d \geq 3$, the one-sector exponent matrix has rows $e_1 + e_2, e_1 + e_3, e_2 + e_3, e_1 + e_4, \ldots, e_1 + e_d$. Its leading block is

$$B_3 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det B_3 = -2,$$

and every later row introduces one new column, so $\operatorname{rank} B_d = d$. The full K3P normalizer matrix is

$$\operatorname{diag}(B_d, B_d, B_d),$$

with rank $3d$ and leading determinant -8. In each sector,

$$a_1 = \sqrt{r_{12}r_{13}/r_{23}}, \quad a_2 = \sqrt{r_{12}r_{23}/r_{13}}, \quad a_3 = \sqrt{r_{13}r_{23}/r_{12}}, \quad a_k = r_{1k}/a_1.$$

8 Fourteen-orbit and 38+2 census

orbit	source	target; port permutation	rank	raw	result
H21-01	S1 (θ_0, r_1)	T80 $(\theta_0, r_1; \texttt{0132})$	$21 \to 21$	4	quartic
H21-02	S1	T80 `0213`	$21 \to 21$	4	rank
H21-03	S1	T80 `0231`	$21 \to 21$	4	quartic
H21-04	S1	T80 `0312`	$21 \to 21$	4	quartic
H21-05	S1	T80 `0321`	$21 \to 21$	2	quartic
H21-06	S1	T80 `1032`	$21 \to 21$	4	quartic
L20-01	S0 (θ_0, r_0)	T822 $(\theta_3, r_1; \texttt{1203})$	$20 \to 24$	2	quartic
L20-02	S0	T822 `1230`	$20 \to 24$	2	rank
L21a-01	S2 (θ_1, r_0)	T822 `0123`	$21 \to 24$	2	quartic
L21a-02	S2	T822 `0132`	$21 \to 24$	2	rank
L21b-01	S3 (θ_1, r_1)	T822 `0123`	$21 \to 24$	2	quartic
L21b-02	S3	T822 `0132`	$21 \to 24$	2	rank
L23-01	S4 (θ_3, r_0)	T822 `0123`	$23 \to 24$	2	rank
L23-02	S4	T822 `0132`	$23 \to 24$	2	quartic

Raw membership sums to 38. Two rank-24 source-5/target-822 sink swaps, with port permutations `0132` and `1023`, are separately quartic-separated. Thus $40 = 38 + 2$, with zero unresolved rows.

9 Two sink-swap records

The two post-quadratic records not absorbed into the fourteen canonical orbits both compare source support 5 with target completion 822. Their port permutations are `0132` and `1023`. Neither transport is a labelled mixed-graph isomorphism. For each record the target pullback of Q_{23} is zero, while the source pullback has 160 terms and is nonzero at a strict rational point of minimum physical margin 7/31. The common source-pullback SHA-256 is abbreviated as `c1f4289be568...0c6f4a`. Consequently neither record yields a new symmetric move or a directed containment.

10 Polynomial separator data

For four-port coordinates, indices $0, \dots, 63$ use the lexicographic conservation-supported order

$$\begin{gathered}0000, 00CC, 00GG, 00TT, 0C0C, 0CC0, 0CGT, 0CTG, 0G0G, 0GCT, 0GG0, 0GTC, \\ 0T0T, 0TCG, 0TGC, 0TT0, C00C, C0C0, C0GT, C0TG, CC00, CCCC, CCGG, CCTT, \\ CG0T, CGCG, CGGC, CGT0, CT0G, CTCT, CTG0, CTTC, G00G, G0CT, G0G0, G0TC, \\ GC0T, GCCG, GCGC, GCT0, GG00, GGCC, GGGG, GGTT, GT0C, GTC0, GTGT, GTTG, \\ T00T, T0CG, T0GC, T0T0, TC0G, TCCT, TCG0, TCTC, TG0C, TGC0, TGGT, TGTG, \\ TT00, TTCC, TTGG, TTTT.\end{gathered}$$

Each body below is a signed sum of eight degree-four monomials. A tuple (i, j, k, l) denotes $q_i q_j q_k q_l$.

body	signed index monomials
$Q_{H,a}$	$+(0,24,44,52)-(0,28,36,56)-(4,24,32,60)+(4,28,40,48)$ $+(8,16,36,60)-(8,20,44,48)-(12,16,40,52)+(12,20,32,56)$
$Q_{H,b}$	$+(0,18,35,49)-(0,19,33,50)-(1,18,32,51)+(1,19,34,48)$ $+(2,16,33,51)-(2,17,35,48)-(3,16,34,49)+(3,17,32,50)$
Q_{20}	$+(0,10,35,61)-(0,11,34,61)+(0,14,41,51)-(0,15,41,50)$ $-(1,10,35,60)+(1,11,34,60)-(1,14,40,51)+(1,15,40,50)$
Q_{21}	$+(0,10,44,55)+(0,11,38,60)-(0,14,40,55)-(0,15,38,56)$ $-(4,10,44,51)-(4,11,34,60)+(4,14,40,51)+(4,15,34,56)$
Q_{23}	$+(0,8,45,53)+(0,9,37,60)-(0,12,37,57)-(0,13,40,53)$ $-(5,8,45,48)-(5,9,32,60)+(5,12,32,57)+(5,13,40,48)$

The per-orbit transport and nonvanishing summary is:

orbit	body	marginal/representative transport	source terms	strict margin	source pullback SHA-256
H21-01	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
H21-03	$Q_{H,b}$	omit 1; retain $(0,2,3)$	192	2/9	`14f7d8b5683d...a8b9da28`
H21-04	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
H21-05	$Q_{H,b}$	omit 1; retain $(0,2,3)$	192	2/9	`14f7d8b5683d...a8b9da28`
H21-06	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
L20-01	Q_{20}	representative `1203`	270	5/23	`34aee883f27a...502a371`
L21a-01	Q_{21}	representative `0123`	270	7/31	`ff97c8b01748...c97521d0`
L21b-01	Q_{21}	representative `0123`	270	5/23	`ff97c8b01748...c97521d0`
L23-02	Q_{23}	representative `0132`	96	2/9	`fd18a85829e2...67bbe338`

All H21 character transports are the identity on the ordered labels C, G, T; only physical ports move. Every target pullback is exactly zero. The two sink swaps use Q_{23}, have 160-term source pullbacks, strict margin 7/31, and common pullback hash `c1f4289be568...0c6f4a`. Full hashes and exact rational source values are in the bound JSON files; abbreviations above are for page fit only.

11 Directed-rank witnesses

The table gives the selected output-row set, source minor columns, strict source rank, target generator upper bound, and target compression. Exact rational determinants and labels are in `input_frozen/k3p_cloud_artifacts/k3p_directed_rank_obstructions.json` (SHA-256 `fa0ac74cde903edc422a90b5e490bd639b2e2b4c758d9d3ec10a794f1a044f42`).

orbit	selected output rows	source columns	gap	target factorization
H21-02	$(3,12,15,20,27,39,$ $40,48,51,60,63)$	$(2,3,4,5,6,7,$ $8,11,14,20,23)$	$11>10$	ten generators $U, V, Z, D, I, A_0, B_0, A, B, \rho$, saturated at $e_{2C}e_{2G}DI \neq 0$
L20-02	$(5,10,15,17,20,27,30,34,$ $39,40,45,51,54,57,60)$	$(3,4,5,6,7,8,9,$ $10,11,12,13,14,18,19)$	$14>12$	ordinary-sunlet compression; omit port 3
L21a-02	$(5,15,17,20,27,39,$ $40,45,51,57,60)$	$(3,4,5,6,7,8,$ $9,10,11,15,16)$	$11>10$	sunlet generators with A_G, B_G absent; omit port 3
L21b-02	same as L21a-02	same	$11>10$	same ten-generator support; omit port 3
L23-01	$(4,8,12,16,20,24,28,32,$ $36,40,44,48,52,56,60)$	$(3,4,5,6,7,8,9,$ $10,11,12,13,14,21,22)$	$14>12$	ordinary-sunlet compression; omit port 2

The exact target minors have ranks $10, 12, 10, 10, 12$, respectively, and the function-field formulas prove these are upper bounds, not merely sampled ranks.

For H21-02, L21a-02, and L21b-02, the source minor uses the complete displayed row set. For L20-02 and L23-01, it uses the displayed set with row 60 deleted. The bound JSON records the determinant-sensitive row order and the exact nonzero rational determinant for every source minor.

12 Restoration census

certificate	minimal first layer	complete redundant forest
displayed quartet	35,758	36,006
K3P tree–sunlet SOS	606	614
K3P multihomogeneous quadratic	148	148
active K3P marginal quartic	56	56
edges/terminal rows	36,568	36,824

The imported graph forest has 997 parents, 36,568 first edges, 32 legacy structural continuation nodes, 256 depth-two edges, and 36,792 legacy leaves, with maximum depth two. K3P quartics terminate all 32 continuations at depth one, so the active minimal proof has 36,568 terminal rows. The depth-two replay remains as a redundant graph-and-algebra audit. There are zero missing/duplicate children, cycles, broken transports, or unresolved rows.

The 56 all-edge K3P quartics split as 24 formerly transported quartic rows plus the 32 legacy continuation nodes, all regenerated and checked in the active K3P coordinate system.

The producer-independent replay passed with canonical payload `91a29973bcd333c05270ffd379427a8c949285f791fd5db299a83c46934e82b1`. The mutation suite rejected all 20 mutations, with payload `f770cecb54b459b3ff11d85b0df780e10c07b8fff92dee553ce0feb91120a9c1`. The sealed manifest payload is `b35346386300bb3f0816a2bbe671280f92f28ee1c5fa2eb9829e6f9fe0863c39`.

13 Probe census

The 176 anchors comprise 143 isomorphisms and 33 triangle relations in 39 canonical classes.

category	one port	two ports
displayed quartet	27,758	511,266
K3P tree–sunlet SOS	99	576
isomorphism	1,915	30,969
triangle	192	1,760
raw rows	29,964	544,571
equality survivors	2,107	32,729
unresolved	0	0

The transport package has 67,741 exact transport rows and 4,379 parent restrictions, with zero incoherent choices. Ordered ledger roots and compressed-ledger SHA-256 values are in `probes/ONE_PORT_PROBE_MANIFEST.json`, `probes/TWO_PORT_PROBE_MANIFEST.json`, and `probes/GLOBAL_TRANSPORT_MANIFEST.json`.

The coherence layer is sealed against the complete replacement K3P restoration theorem by `probes/RESTORATION_MANIFEST.json`, schema `k3p-restoration-manifest-v2`, status `PASS`, with canonical payload `8c7eeeb77c09a64925fd604358b5660decca4699bd7a2fe04a0a52469a740fa7`. This binding distinguishes the 36,568 active K3P terminal rows from the 36,792 legacy/full-forest leaves and excludes historical K2P algebra.

14 Krawczyk certificate summary

The equality slice has 15 scaled pivot variables, 49 frozen direct parameters, and rational box radius 10^{-50}. The fifteen exact sparse equations have term counts

$$(5, 5, 4, 4, 3, 3, 7, 5, 4, 2, 5, 3, 7, 4, 3).$$

The maximum scaled residual at the rational center is approximately $3.02838161 \times 10^{-89}$. The exact center Jacobian determinant is approximately $-3.05693250 \times 10^{-2}$. With exact $Y = J_F(y_0)^{-1}$,

$$\max_j \frac{|K_j - y_{0,j}|}{10^{-50}} = 9.74099938 \times 10^{-41}, \qquad \|I - YJ_F(X)\|_\infty \le 8.07702308 \times 10^{-47}.$$

Both inequalities are rigorous rational interval bounds. The first is a strict Krawczyk inclusion; the second gives uniqueness inside the selected slice.

15 Sharpness rank witnesses

map	selected local columns	point determinant (decimal summary)	uniform Neumann bound
W	$(0, 1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18)$	$1.30773104 \times 10^{-336}$	$1.54315210 \times 10^{-45}$
W'	$(0, 1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17)$	$-6.74132416 \times 10^{-325}$	$4.58271952 \times 10^{-45}$

The smallest certified edge-eigenvalue lower bounds are approximately 4.96448×10^{-10} and 1.39520×10^{-9}; the smallest continuous-time lower bounds are approximately 4.96448×10^{-10} and 1.39520×10^{-9}. Exact rational intervals for every edge, transition entry, inheritance probability, and minor are stored in the 9.9 MB Krawczyk certificate.

The exhaustive topology censuses are $W : (5, 2, 3)$, $W' : (7, 2, 5)$, and outer collision reference $(7, 0, 7)$.

16 Cherry determinant and all-n extension

For $h = C, G, T$, use $R_h = u_h/v_h$ and $P_h = u_h v_h$. Direct formal Laurent differentiation gives

$$\det \frac{\partial(R_C, P_C, R_G, P_G, R_T, P_T)}{\partial(u_C, v_C, u_G, v_G, u_T, v_T)} = \frac{8u_C u_G u_T}{v_C v_G v_T}.$$

At $u = (2/5, 4/9, 3/7)$, $v = (3/7, 5/11, 4/9)$, this is $176/25$. The smallest exact strict physical margins at these two spectra are $83/630$ and $367/2772$. The positive inverse is $u_h = \sqrt{R_h P_h}$, $v_h = \sqrt{P_h/R_h}$. Each cherry adds exactly six dimensions, yielding $15 + 6(n - 3) = 6n - 3$.

17 Clean-room architecture and H21-01 repair

The historical clean-room verifier is preserved at SHA-256 `ee5e29a2cd795d9389e8e1257ebdb9eeaa4256fb5d03e0 7f230bf82ba555ef91`.

It reproduces the H21-01 failure. The defect was mathematical, not a change in the expected orbit: it used rooted-DAG automorphisms instead of root-suppressed semi-directed mixed-graph automorphisms and did not conjugate the base target automorphism into the displayed frame. The corrected replay:

- binds five frozen inputs by hard-coded SHA-256 before parsing;
- reconstructs source and target incoming roles, repairs, physical ports, and port permutations from literal graphs;
- computes all seven H21 double cosets and the six nonisomorphic ones;
- replays all 38 raw Fourier-coordinate transports;
- binds each rank label to a square minor, selected rows and columns; and
- independently derives every target function-field rank upper bound.

The corrected verifier has zero Python assertion nodes and refuses optimized mode before reading inputs.

18 Mutation summary

The integrated primary verifier passes 28 of 28 active evidence entries. The final classification gate then reports `CERTIFIED_K3P_SAME` with no remaining mathematical gates. Its own fail-closed suite rejects 16 of 16 integrated mutations. The active component suites also include:

component	rejected	representative mutations
H21/fourteen-orbit gate	10	port/incoming changes, weakened rank bounds, nonsquare minor, selected-row change, optimized mode, input byte change
H21 adversarial re-audit	25	raw coverage, target conjugation, coordinate transport, graph-role and exact-identity corruptions
cross-bridge local theorem	34	missing target, partition overlap, changed single/pair/cyclic identities, record-43/60 bindings
global cut transfer	30+32	circular dependency, crossing compatibility, side-blob physicality, target openness, tree-coloring counterexamples
integrated cut claim boundary	12	universal-pointwise substitution, claim-scope drift, circular dependencies, and ordinary/optimized modes
global analytic infrastructure	16	$C = T$, state permutation, missing domain inequality, rank-15 triangle, orientation-specific context
probe coherence	17	missing anchor/probe/parent, reversed order, broken transport, new/inconsistent triangle, $C = T$, optimized mode
sharpness	18	rooting census, graph status, Krawczyk data, interval rank, physical margins, cherry determinant and persistence
restoration	20	missing/duplicated child, wrong parent, K2P algebra, altered SOS/quadratic/quartic, broken marginal transport
final integrated K3P-SAME gate	16	universal-pointwise substitution, rank-15 triangle, proper directed containment, stale restoration binding, K2P leakage, reversed CT transfer, submission-ready and optimized-mode bypasses

Each campaign records the expected and observed diagnostic; a stale hash alone is not accepted as the intended mathematical rejection when the suite is designed to rebind mutated inputs.

19 Reproduction commands

From the project root, the component-level commands used by this draft are:

```
.venv/bin/python reproducibility/verify_k3p_same_classification.py
.venv/bin/python reproducibility/test_k3p_same_classification_mutations.py

bash reproducibility/verify_primary.sh
bash clean_room/verify_clean_room.sh

.venv/bin/python \
  cut_recovery/strong_crossbridge/final_certificate/build_final_certificate.py
.venv/bin/python \
  cut_recovery/strong_crossbridge/final_certificate/verify_final_certificate.py
.venv/bin/python \
  cut_recovery/strong_crossbridge/global_transfer/verify_release.py

.venv/bin/python restoration/regenerate_k3p_restoration.py --resume
.venv/bin/python restoration/verify_k3p_restoration.py
.venv/bin/python restoration/test_k3p_restoration_mutations.py

.venv/bin/python probes/regenerate_k3p_probes.py
.venv/bin/python probes/verify_k3p_probes.py
.venv/bin/python probes/test_k3p_probe_mutations.py

.venv/bin/python global_infrastructure/generate_global_infrastructure.py
.venv/bin/python global_infrastructure/verify_global_infrastructure.py
.venv/bin/python global_infrastructure/test_global_infrastructure_mutations.py
```

```
.venv/bin/python sharpness/independent_krawczyk_replay.py
.venv/bin/python sharpness/independent_topology_alln_replay.py
```

The first two commands are the sealed final mathematical classification gate and its 16-mutation fail-closed suite. The longer producer commands are listed only for complete regeneration; no release-archive or submission gate is claimed here.

20 Artifact hashes

artifact	SHA-256
model_domain/primary_exact_evidence.json	ac21e4f795537f251377411841d670c1bad4ce06a69ed2 4f596252c11cb7afb6
three_port/primary_exact_evidence.json	79cd462c3afb88f7cf1f898febbb2ad1a7a701c7bb05a5 8a833ed96736fdc5de
bridge_fibre/K3P_BRIDGE_FIBRE_CERTIFICATE.json	bdd8e8f74a896b5031a50cad9ac97a82a03f4f821f2dc7 93cfb386415b1df532
marginals/K3P_MARGINAL_SUBMERSION_CERTIFICATE.json	f85f918a770af7482abfe914bc93a304ce23c248906264 3036b7a742ab09ef7f
four_port_atlas/primary_exact_evidence.json	f4e02c74e4b4f9fbb063b13da65033a46ef45f2eecaad2 b9efc1895b63afadab
clean_room/H21_01_TRANSPORT_AUDIT.json	b8a1c52a01ce0bb1b50a8ffaf78f5a300be52fde976671 b2b1cc91e8fe2d7229
triangle_h14/K3P_H14_CONTEXT_CERTIFICATE.json	082f24cd26be08f248ddc6d0cf804059a27ad96477b06d f0c56f8a89818bb9b9
cut_recovery/strong_crossbridge/final_certificate/STRONG_CROSSBR IDGE_FINAL_CERTIFICATE.json	643c29780219a538a5a127341ea91363aa5899d6f0f6fa 1dac034889a7fdf06b
cut_recovery/strong_crossbridge/global_transfer/THEOREM_MANIFEST .json	00021da5e23726fa6a34e0c024b0703bb79e2dbcdecb58 affe559e01746c7895
global_infrastructure/GLOBAL_INFRASTRUCTURE_MANIFEST.json	5c9450b51eaa8bc256523932c2cd7d9025c38230e2e2ce 894b9f947086ae06b0
reproducibility/primary_gate_report.json	19990e65f17834520c676206bc70f2f80fa37b191b1382 1536fe366e78de1c45
reproducibility/K3P_SAME_CLASSIFICATION_GATE_REPORT.json	2dc2371a185597571e998fe6a271c9f9769e2561f372f1 4e13328bffea794954
reproducibility/K3P_SAME_CLASSIFICATION_MUTATION_REPORT.json	8457a1541ac2b41f5bb4f91cfec0111deec5d282b753c3 233c581e31fb057f8d
restoration/RESTORATION_MANIFEST.json	a95976496e3f0a5ad289db5dabac688dd1630082eeaaff 2f484c113882a5bf8e
restoration/K3P_RESTORATION_THEOREM_REPORT.md	9c3ea74acdc5dc4eff24d50c9f2db845215bde6b8276c7 2faf0a6d150f4bf7f0
restoration/K3P_RESTORATION_INDEPENDENT_VERIFICATION.json	925fc8855c92b3b625287347a0e6ee0f8e441a58ac09e1 50f7521b92b8d3e1fa
restoration/K3P_RESTORATION_MUTATION_CERTIFICATE.json	5fac1f616e8bd8bb1c2290debad5d624cb54d14caeecaa 601550e8fb19dcac2a
probes/K3P_PROBE_COHERENCE_CERTIFICATE.json	45411c6f4924292b975f2620ec9f32c272628378b0f2b3 c3b746abbc8ba4e2d8
probes/RESTORATION_MANIFEST.json	b6f28a6cd44711cffd939f9e8f72c4ef61ac2a89c265f8 5535687a370fbe12ea
sharpness/K3P_SHARPNESS_KRAWCZYK_CERTIFICATE.json	8187174b3e0c0b3a0a55fa32595c211811c357dc223ada 7a74b3033f7cae3941
sharpness/K3P_SHARPNESS_TOPOLOGY_ALL_N_CERTIFICATE.json	aa0883777445541398bed943881405530c9b5bb4f0451 794a4b56b289beabc7

These are the frozen active hashes at this revision, including the final mathematical classification gate. A later release manifest must supersede this typeset table if packaging changes any byte; no release archive or submission package is asserted here.